Grade 10 – Term 1
Decision-Making Under Uncertainty
Exam Solutions – Version 2D

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Learning objective

Compare decision criteria and their limitations, and construct and interpret decision trees for staged filtering under uncertainty.

Criteria assessment

This exam assesses C8 and C9, each worth 6 points. A **minimum of 4 points per criterion** is required to pass. Show all relevant calculations and justify your answers. Partial credit may be awarded for correct procedures despite arithmetic errors; answers without demonstrated procedures receive no credit. A 1-point penalty per criterion applies if the student's name is missing or the assessment is submitted late. C10 is assessed separately.

Question	C8	C9
1	3	–
2	3	–
3	–	2
4	–	2
5	–	2
Total	**6**	**6**

Evaluated criteria

C8 Compares the Maximax, Maximin, Minimax Regret, and EMV/Bayes decision criteria by contrasting their information requirements, risk orientations, recommended alternatives, and limitations.

C9 Constructs and interprets decision trees for staged filtering rules, applying decision criteria in the specified order to identify the final alternative.

C10 Evaluates economic and financial decision-making scenarios by selecting and justifying the most appropriate decision criterion according to the information available, risk attitude, and decision context.

1 Camera Rental Plans [C8-3]

A camera rental studio selects its inventory under uncertain demand. Profits are in thousand USD.

Alternative	High	Normal	Low
$P(S_j)$	0.3	0.4	0.3
Essential (A_1)	95	88	62
Shared (A_2)	135	148	32
Extensive (A_3)	235	46	-38
Modular (A_4)	165	125	-18

Decision statistics (Regret = maximum regret):

Alt.	Max	Min	Regret	EMV
A_1	95	62	140	82.3
A_2	148	32	100	109.3
A_3	235	-38	102	77.5
A_4	165	-18	80	94.1

(a) Using the original camera rental payoffs, identify the alternative recommended by Maximax, Maximin, Minimax Regret, and EMV/Bayes, and report each criterion's deciding statistic value. **[1]**

(b) A subsidy s (integer, $0 \leq s \leq 90$) increases the low-demand payoffs of A_3 and A_4 by s. An alternative is eligible only if all its payoffs are nonnegative. Determine each camera rental plan's eligibility based on the value of s. **[0.5]**

(c) Among eligible camera rental plans, find the smallest allowed subsidy s at which the Maximin recommendation changes from A_1 to A_4 alone. Identify any tie at the threshold and distinguish it from a strict change. Respect the integer domain $0 \leq s \leq 90$; if no smallest value exists, state the threshold and the condition for a change. **[1.5]**

(a) Initial recommendations [1]

Criterion	Recommended alternative(s)	Deciding statistic and value
Maximax	A_3	Row maximum: 235
Maximin	A_1	Row minimum: 62
Minimax Regret	A_4	Maximum regret: 80
EMV/Bayes	A_2	EMV: 109.3

All deciding values are in thousand USD. Maximax pursues the greatest upside; Maximin protects the worst outcome; Minimax Regret limits missed profit; EMV uses the forecast probabilities to maximize expected profit. Regret benchmarks use all four original alternatives, with state bests $(235, 148, 62)$; no eligibility restriction applies in (a).



(b) Subsidy and eligibility [0.5]

The updated payoff table is:

Alternative	High	Normal	Low
$P(S_j)$	0.3	0.4	0.3
Essential (A_1)	95	88	62
Shared (A_2)	135	148	32
Extensive (A_3)	235	46	$-38+s$
Modular (A_4)	165	125	$-18+s$

The updated decision statistics are:

Alt.	Max	Min	Regret	EMV
A_1	95	62	140	82.3
A_2	148	32	100	109.3
A_3	235	$\min(46, -38+s)$	$\max(102, 100-s)$	$77.5+0.3s$
A_4	165	$-18+s$	$\max(70, 80-s)$	$94.1+0.3s$

An alternative is eligible only if all its payoffs are nonnegative. For integer subsidies $0 \le s \le 90$:

Alternative	Eligibility condition	Eligible subsidies
A_1	$62 \ge 0$	$0 \le s \le 90$
A_2	$32 \ge 0$	$0 \le s \le 90$
A_3	$-38+s \ge 0$	$38 \le s \le 90$
A_4	$-18+s \ge 0$	$18 \le s \le 90$

Thus, the eligible sets are:

$$\begin{aligned} 0 \le s \le 17: &\quad \{A_1, A_2\}, \\ 18 \le s \le 37: &\quad \{A_4, A_1, A_2\}, \\ 38 \le s \le 90: &\quad \{A_3, A_4, A_1, A_2\}. \end{aligned}$$

(c) Maximin change under subsidy s [1.5]

The minimum payoffs are

$$m_1 = 62, \quad m_2 = 32, \quad m_3(s) = \min(46, -38 + s), \quad m_4(s) = -18 + s.$$

Since $m_3(s) \le 46 < 62 = m_1$ and $m_2 = 32 < 62$ throughout the allowed range, only A_4 can challenge A_1.

For A_4 to beat A_1:

$$-18 + s > 62 \quad \Longleftrightarrow \quad s > 80.$$

Thus, A_1 is the sole Maximin choice for $0 \le s \le 79$; at $s = 80$, A_4 and A_1 tie with minimum payoff 62; and A_4 is the sole Maximin choice for $81 \le s \le 90$. The tie at $s = 80$ first changes the set of recommended plans; a strict change to A_4 alone requires $s > 80$. The integer domain contains a smallest value satisfying this condition. Therefore, the smallest allowed integer subsidy giving a strict change is

$$\boxed{s = 81}.$$

At $s = 81$, all four alternatives are eligible and their minimum payoffs are $62, 32, 43, 63$, respectively. Hence A_4 wins uniquely with a guaranteed profit of 63 thousand USD.

2 Outdoor Cinema Refreshments [C8-3]

A cinema manager selects a refreshment kiosk under uncertain audience size. Profits are in hundred USD.

Alternative	High	Low
$P(S_j)$	0.5	0.5
Portable (C_1)	92	60
Full-service (C_2)	150	16
Deluxe (C_3)	176	-16
Fixed (C_4)	136	46

Decision statistics (Regret = maximum regret):

Alternative	Max	Min	Regret	EMV
C_1	92	60	84	76
C_2	150	16	44	83
C_3	176	-16	76	80
C_4	136	46	40	91

(a) Using the original refreshment kiosk payoffs, identify the alternative recommended by Maximax, Maximin, Minimax Regret, and EMV/Bayes, and report each criterion's deciding statistic value. **[1]**

(b) A sponsor replaces C_1's large-audience payoff of 92 with g (integer, $92 \leq g \leq 156$). An alternative is eligible only if all its payoffs are at least 38. Determine each refreshment kiosk's eligibility based on the value of g. **[0.5]**

(c) Among eligible refreshment kiosks, find the smallest allowed sponsored payoff g at which the Minimax Regret recommendation changes from C_4 to C_1 alone. Recalculate regret benchmarks using only eligible kiosks. Identify any tie at the threshold and distinguish it from a strict change. Respect the integer domain $92 \leq g \leq 156$; if no smallest value exists, state the threshold and the condition for a change. **[1.5]**

(a) Initial recommendations [1]

Criterion	Recommended alternative(s)	Deciding statistic and value
Maximax	C_3	Row maximum: 176
Maximin	C_1	Row minimum: 60
Minimax Regret	C_4	Maximum regret: 40
EMV/Bayes	C_4	EMV: 91

All values are in hundred USD. Regret benchmarks use all four original kiosks, with state bests $(176, 60)$; no eligibility restriction applies in (a). Only EMV uses the probabilities; the other criteria compare upside, downside, or missed profit.

(b) Sponsored payoff and eligibility [0.5]

The updated payoff table is:

Alternative	High	Low
$P(S_j)$	0.5	0.5
Portable (C_1)	g	60
Full-service (C_2)	150	16
Deluxe (C_3)	176	-16
Fixed (C_4)	136	46

The updated decision statistics, before applying eligibility restrictions, are:

Alt.	Max	Min	Regret	EMV
C_1	g	60	$176 - g$	$0.5g + 30$
C_2	150	16	44	83
C_3	176	-16	76	80
C_4	136	46	40	91

An alternative is eligible only if both payoffs are at least 38. For integers $92 \leq g \leq 156$:

Alternative	Eligibility condition	Eligible values
C_1	$g \geq 38,\ 60 \geq 38$	$92 \leq g \leq 156$
C_2	$16 < 38$	None
C_3	$-16 < 38$	None
C_4	$136 \geq 38,\ 46 \geq 38$	$92 \leq g \leq 156$

The eligible set is always $\{C_1, C_4\}$.

(c) Minimax Regret change under sponsored payoff g [1.5]

Recalculate regrets using only these alternatives, with state benchmarks

$$b_H = \max\{g, 136\}, \qquad b_L = 60.$$

Their maximum regrets are

$$R(C_1) = \max\{136 - g, 0\},$$
$$R(C_4) = \max\{g - 136, 14\}.$$

For $92 \leq g \leq 136$, these become

$$R(C_1) = 136 - g, \qquad R(C_4) = 14.$$

For C_1 to be the sole Minimax Regret choice:

$$136 - g < 14 \quad \Longleftrightarrow \quad g > 122.$$

At $g = 122$, both alternatives have maximum regret 14. Thus, C_4 is the sole Minimax Regret choice for $92 \leq g \leq 121$; at $g = 122$, the recommendation becomes the tie $\{C_1, C_4\}$; and C_1 is the sole choice for $123 \leq g \leq 156$. For $g > 136$, C_1 has zero regret and C_4 has regret at least 14, so no further change occurs. The tie first changes the set of recommended kiosks, but a strict change to C_1 alone requires $g > 122$. The integer domain contains a smallest value satisfying this condition. Therefore, the smallest allowed integer giving a strict change is

$$g = 123\ .$$

At this value, the eligible payoff table is:

Alternative	High	Low
C_1	123	60
C_4	136	46

The corresponding maximum regrets are

$$R(C_1) = 13, \qquad R(C_4) = 14.$$

Thus, C_1 wins uniquely with $g = 123$, corresponding to \$12,300 in large-audience profit.

3 Garden Centre Stock [C9-2]

A garden centre selects a seedling stock plan D_1–D_6. Net profits are in hundred USD. The decision statistics are summarized below:

Alternative	Max	Min	Regret	EMV
D_1	120	56	48	102.4
D_2	136	40	32	113.6
D_3	104	72	64	92.8
D_4	152	24	48	113.6
D_5	88	56	80	76.8
D_6	168	8	64	113.6

Apply these stages in order:

1. EMV/Bayes: Best 3.
2. Maximax: Best 2.
3. Maximin: Final selection.

Illustrate each filtering stage, showing survivors and eliminations.

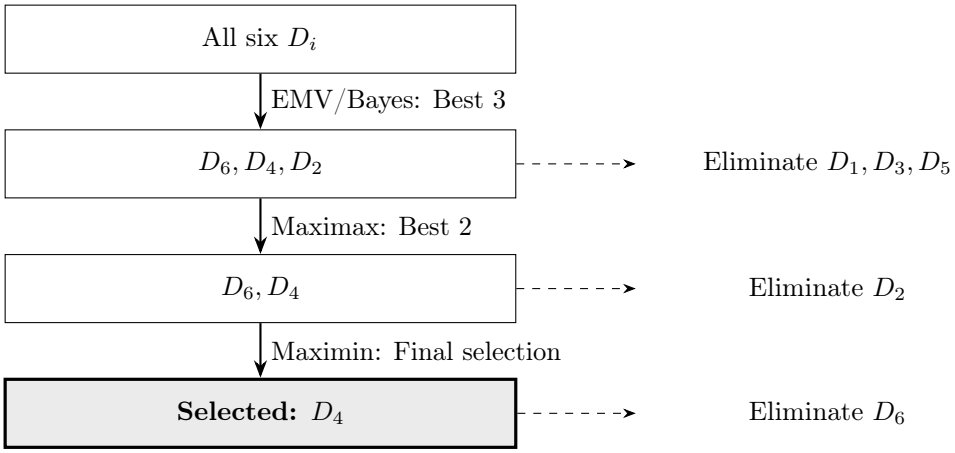
Supporting filtering table

Step	Rule	Survivors (relevant values)	Eliminated (values)
0	Initial alternatives	$D_6, D_4, D_2, D_1, D_3, D_5$	—
1	EMV/Bayes: Best 3	$D_6(113.6)$, $D_4(113.6)$, $D_2(113.6)$	$D_1(102.4)$, $D_3(92.8)$, $D_5(76.8)$
2	Maximax: Best 2	$D_6(168)$, $D_4(152)$	$D_2(136)$
3	Maximin: Final selection	**Selected:** $D_4(24)$	$D_6(8)$

The highest EMV, 113.6, is shared by D_6, D_4, D_2; all three fill the Best 3 places. Maximax retains D_6, D_4 with maxima 168 and 152, eliminating D_2 with 136. Maximin selects D_4 because $24 > 8$. There are no cutoff or final ties beyond the shared first-stage EMV.



Filtering tree and interpretation



The garden centre selects D_4: it survives the EMV and Maximax filters, then its minimum profit of 24 hundred USD exceeds D_6's 8.

4 Climbing Gym Expansion [C9-2]

A climbing gym selects an expansion plan F_1–F_6. Net profits are in thousand USD. The decision statistics are summarized below:

Alternative	Max	Min	Regret	EMV
F_1	253	71	65	184.1
F_2	201	123	117	172.4
F_3	227	97	91	195.8
F_4	318	6	117	164.6
F_5	175	110	143	149
F_6	279	45	78	173.7

Apply these stages in order:

1. Minimax Regret: Best 5.
2. EMV/Bayes: Best 4.
3. Maximax: Best 2.
4. Maximin: Final selection.

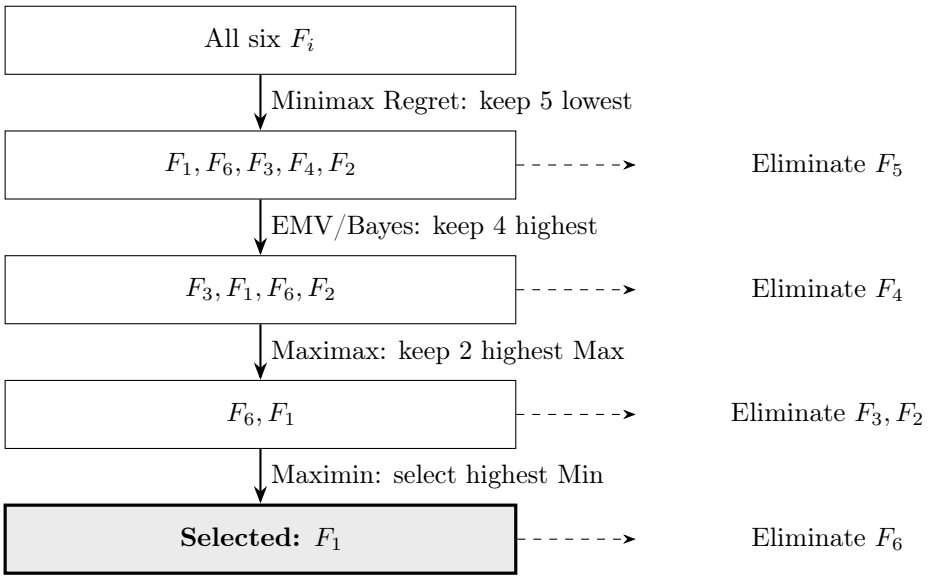
Illustrate each filtering stage, showing survivors and eliminations.

Supporting filtering table

Step	Rule	Survivors (relevant values)	Eliminated (values)
0	Initial alternatives	$F_4, F_6, F_1, F_3, F_2, F_5$	—
1	Minimax Regret: keep 5 lowest	$F_1(65)$, $F_6(78)$, $F_3(91)$, $F_4(117)$, $F_2(117)$	$F_5(143)$
2	EMV/Bayes: keep 4 highest	$F_3(195.8)$, $F_1(184.1)$, $F_6(173.7)$, $F_2(172.4)$	$F_4(164.6)$
3	Maximax: keep 2 highest Max	$F_6(279)$, $F_1(253)$	$F_3(227)$, $F_2(201)$
4	Maximin: select highest Min	**Selected:** $F_1(71)$	$F_6(45)$

At the first cutoff, F_4 and F_2 both have regret 117, and both survive. Later cutoffs have no ties.

Filtering tree and interpretation



The climbing gym selects F_1: after the regret, EMV, and upside filters, its minimum profit of 71 thousand USD exceeds F_6's 45.

5 Art Materials Orders [C9-2]

An art materials distributor selects an order plan H_1–H_6. Net profits are in hundred USD. The decision statistics are summarized below:

Alternative	Max	Min	Regret	EMV
H_1	142	88	81	122.2
H_2	160	70	63	136.6
H_3	196	34	54	123.1
H_4	124	79	99	106
H_5	223	7	81	116.8
H_6	178	52	45	130.3

Apply these stages in order:

1. Maximin: Best 4.
2. Minimax Regret: Best 3.
3. Maximax: Best 2.
4. EMV/Bayes: Final selection.

Illustrate each filtering stage, showing survivors and eliminations.

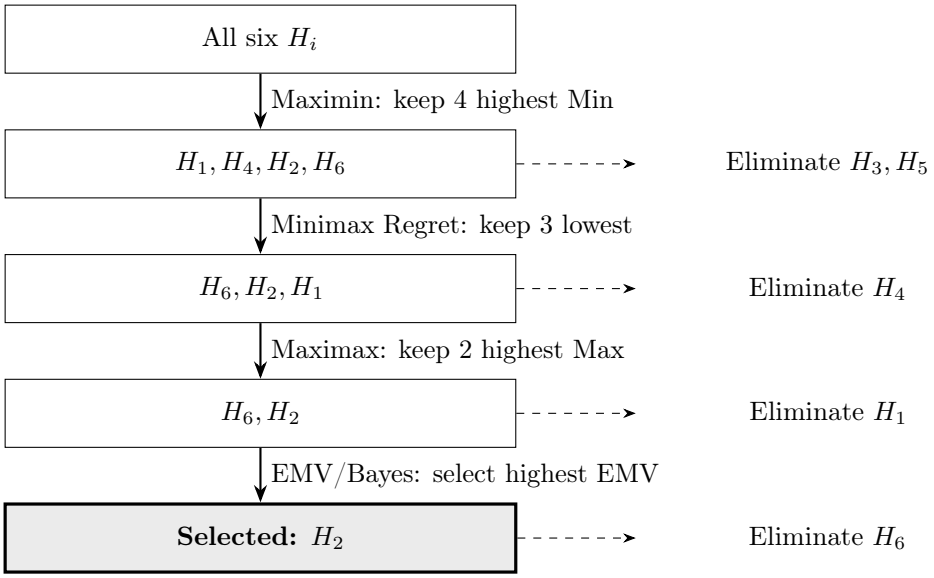
Supporting filtering table

Step	Rule	Survivors (relevant values)	Eliminated (values)
0	Initial alternatives	$H_5, H_3, H_6, H_2, H_1, H_4$	—
1	Maximin: keep 4 highest Min	$H_1(88)$, $H_4(79)$, $H_2(70)$, $H_6(52)$	$H_3(34)$, $H_5(7)$
2	Minimax Regret: keep 3 lowest	$H_6(45)$, $H_2(63)$, $H_1(81)$	$H_4(99)$
3	Maximax: keep 2 highest Max	$H_6(178)$, $H_2(160)$	$H_1(142)$
4	EMV/Bayes: select highest EMV	**Selected:** $H_2(136.6)$	$H_6(130.3)$

The cutoffs are Min 52, maximum regret 81, and Max 160; the final EMV comparison is 136.6 versus 130.3. There are no cutoff or final ties.



Filtering tree and interpretation



The art materials distributor selects H_2: after the downside, regret, and upside filters, its EMV of 136.6 hundred USD exceeds H_6's 130.3.

C10: Separate activity reminder (unscored)

Complete the individual **Interactive Decision Webpage: C10 Practice** activity and submit the deployed webpage URL. See the separate activity for instructions.

C10 example solution.

C5 catch Up

Camera Rental Plans (Question 1). Apply Minimax Regret to the original payoff table.

A_1 Essential, A_2 Shared, A_3 Extensive, A_4 Modular.
S_1 High, S_2 Normal, S_3 Low.
Profits and regrets are in thousands of USD.
State bests are $\boldsymbol{b} = (235, 148, 62)$.
Regret is $r_{ij} = b_j - P_{ay}(A_i, S_j)$.

Regret	S_1	S_2	S_3	Max.
A_1	$235 - 95 = 140$	$148 - 88 = 60$	$62 - 62 = 0$	140
A_2	$235 - 135 = 100$	$148 - 148 = 0$	$62 - 32 = 30$	100
A_3	$235 - 235 = 0$	$148 - 46 = 102$	$62 - (-38) = 100$	102
A_4	$235 - 165 = 70$	$148 - 125 = 23$	$62 - (-18) = 80$	80

$$\min\{140, 100, 102, 80\} = 80 \implies A_4 \text{ (Modular)} \quad \text{Maximum regret } 80$$

6 Summarized Solutions [C8–C9]

6.1 Camera Rental Plans C8

Values in thousands of USD.

(a) Original payoffs

Criterion	Choice	Value
Maximax	A_3	235
Maximin	A_1	62
Minimax Regret	A_4	80
EMV/Bayes	A_2	109.3

(b) Eligibility

Alternative	Eligible range of s
A_1	$0 \le s \le 90$
A_2	$0 \le s \le 90$
A_3	$38 \le s \le 90$
A_4	$18 \le s \le 90$

(c) Maximin

- Compare eligible minima $m_1 = 62$ and $m_4 = s - 18$. Both $m_2 = 32$ and $m_3 \le 46$ are below m_1.
- A_1 and A_4 tie at $s = 80$ with minimum 62.
- First sole win for A_4 at $s = 81$, with minimum 63.

6.2 Outdoor Cinema Refreshments C8

Values in hundreds of USD.

(a) Original payoffs

Criterion	Choice	Value
Maximax	C_3	176
Maximin	C_1	60
Minimax Regret	C_4	40
EMV/Bayes	C_4	91

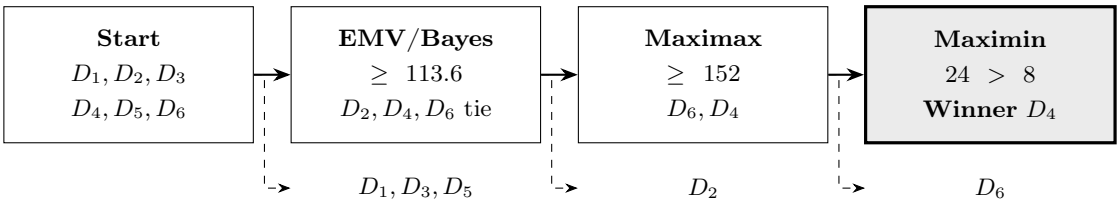
(b) Eligibility

Alternative	Eligible range of g
C_1	$92 \le g \le 156$
C_2	None
C_3	None
C_4	$92 \le g \le 156$

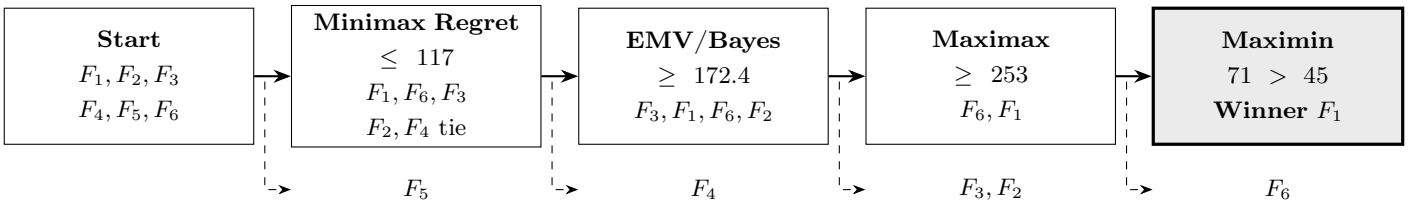
(c) Minimax Regret

- Eligible state bests (high, low) from C_1, C_4 are $(\max(g, 136), 60)$.
- Maximum regrets are $R_1 = \max(136 - g, 0)$ and $R_4 = \max(g - 136, 14)$.
- C_1 and C_4 tie at $g = 122$ with regret 14.
- First sole win for C_1 at $g = 123$, where $13 < 14$.

6.3 Garden Centre Stock C9



6.4 Climbing Gym Expansion C9



6.5 Art Materials Orders C9

